PRACTICAL 1 - OSPF, SSH, NTP AND SYSLOG

!!!Scripts for R1!!!!

```
enable
configure terminal

interface s0/0/0
ip ospf message-digest-key 1 md5 MD5pa55
exit

router ospf 1
area 0 authentication message-digest
exit

service timestamps log datetime msec
logging 192.168.1.6

ntp authentication-key 1 md5 NTPpa55
ntp authenticate
ntp trusted-key 1
ntp server 192.168.1.5
ntp update-calendar
end
```

!!!Scripts for R2!!!!

```
enable
configure terminal

interface s0/0/0
ip ospf message-digest-key 1 md5 MD5pa55
exit

interface s0/0/1
ip ospf message-digest-key 1 md5 MD5pa55
exit

router ospf 1
area 0 authentication message-digest
exit

service timestamps log datetime msec
logging 192.168.1.6

ntp authentication-key 1 md5 NTPpa55
ntp authenticate
ntp trusted-key 1
ntp server 192.168.1.5
ntp update-calendar
end
```

!!!Scripts for R3!!!!

```
enable
configure terminal

interface s0/0/1
ip ospf message-digest-key 1 md5 MD5pa55
exit

router ospf 1
area 0 authentication message-digest
exit

service timestamps log datetime msec
logging 192.168.1.6

ntp authentication-key 1 md5 NTPpa55
ntp authenticate
ntp trusted-key 1
ntp server 192.168.1.5
ntp update-calendar

ip domain-name ccnasecurity.com
username SSHadmin privilege 15 secret ciscosshpa55

crypto key generate rsa modulus 1024

ip ssh version 2
ip ssh time-out 90
ip ssh authentication-retries 2

line vty 0 4
login local
transport input ssh
exit
end
```

PRACTICAL 2 - AAA AUTHENTICATION

!!!Script for R1!!!

```
!! PART 1
enable
configure terminal
username Admin1 secret admin1pa55
aaa new-model
aaa authentication login default local
line console 0
login authentication default
exit
!!PART 2
ip domain-name ccnasecurity.com
crypto key generate rsa
1024

aaa authentication login SSH-LOGIN local

line vty 0 4
login authentication SSH-LOGIN
transport input ssh
exit
end
```

!!!Script for R2!!!

```
enable
configure terminal
username Admin2 secret admin2pa55

tacacs-server host 192.168.2.2
tacacs-server key tacacspa55

aaa new-model
aaa authentication login default group tacacs+ local

line console 0
login authentication default
exit
end
```

!!!Script for R3!!!

```
enable
configure terminal
username Admin3 secret admin3pa55
radius-server host 192.168.3.2
radius-server key radiuspa55
aaa new-model
aaa authentication login default group radius local
line console 0
login authentication default
Exit > end
```

PRACTICAL 3 - EXTENDED ACL

R1 – ACL 100 (FTP + ICMP)

```
enable
configure terminal
access-list 100 permit tcp 172.22.34.64 0.0.0.31 host 172.22.34.62 eq ftp
access-list 100 permit icmp 172.22.34.64 0.0.0.31 host 172.22.34.62
interface g0/0
ip access-group 100 in
exit
end
```

R1 – Named ACL HTTPONLY

```
configure terminal
ip access-list extended HTTPONLY
permit tcp 172.22.34.96 0.0.0.15 host 172.22.34.62 eq www
permit icmp 172.22.34.96 0.0.0.15 host 172.22.34.62
exit
interface g0/1
ip access-group HTTPONLY in
exit
end
```

RTA – ACL 199

```
enable
configure terminal
access-list 199 permit tcp 10.101.117.32 0.0.0.15 10.101.117.0 0.0.0.31 eq 22
access-list 199 permit icmp any any
interface g0/2
ip access-group 199 out
exit
end
```

PRACTICAL 4 - IP ACL TO MITIGATE ATTACKS

!!!Script for R1!!!

```
enable
configure terminal

! Secure Router Access (VTY Protection)
access-list 10 permit host 192.168.3.3
line vty 0 4
access-class 10 in
exit

! ACL 120 – Permit Required Services & Block HTTPS
access-list 120 permit udp any host 192.168.1.3 eq domain
access-list 120 permit tcp any host 192.168.1.3 eq smtp
access-list 120 permit tcp any host 192.168.1.3 eq ftp
access-list 120 deny tcp any host 192.168.1.3 eq 443
access-list 120 permit tcp host 192.168.3.3 host 10.1.1.1 eq 22

! Allow ICMP Replies
access-list 120 permit icmp any any echo-reply
access-list 120 permit icmp any any unreachable
access-list 120 deny icmp any any

! Allow Remaining Traffic
access-list 120 permit ip any any

! Apply ACL
interface s0/0/0
ip access-group 120 in
Exit > end
```

!!!Script for R2!!!

```
enable
configure terminal

access-list 10 permit host 192.168.3.3
line vty 0 4
access-class 10 in
exit>end
```

!!!Script for R3!!!

```
enable
configure terminal

! Secure Router Access
access-list 10 permit host 192.168.3.3
line vty 0 4
access-class 10 in
exit

! Bogon / RFC1918 Filtering – ACL 100
```

```
access-list 100 permit tcp 10.0.0.0 0.255.255.255 host 192.168.3.3 eq 22
access-list 100 deny ip 10.0.0.0 0.255.255.255 any
access-list 100 deny ip 172.16.0.0 0.15.255.255 any
access-list 100 deny ip 192.168.0.0 0.0.255.255 any
access-list 100 deny ip 127.0.0.0 0.255.255.255 any
access-list 100 deny ip 224.0.0.0 15.255.255.255 any
access-list 100 permit ip any any

interface s0/0/1
ip access-group 100 in
exit

! Anti-Spoofing ACL 110
access-list 110 permit ip 192.168.3.0 0.0.0.255 any

interface g0/1
ip access-group 110 in
exit>end

! Final Verification Commands (All Routers)
show access-lists
show ip interface brief
show running-config
write memory
```

PRACTICAL 5 - IPv6 ACL

!!!Scripts for R1!!!

```
enable
conf t
ipv6 unicast-routing

ipv6 access-list BLOCK_HTTP
 deny tcp any host 2001:DB8:1:30::30 eq www
 deny tcp any host 2001:DB8:1:30::30 eq 443
 permit ipv6 any any
exit

interface g0/1
ipv6 traffic-filter BLOCK_HTTP in
exit>end
```

!!!Scripts for R2!!!

```
enable
conf t
ipv6 unicast-routing
end
```

!!!Scripts for R3!!!

```
enable
conf t
ipv6 unicast-routing

ipv6 access-list BLOCK_ICMP
 deny icmp any any
 permit ipv6 any any
exit

interface g0/0
 ipv6 traffic-filter BLOCK_ICMP out
exit>end
```

!!! PC1 > Desktop > IP Config

IPv6 Address: 2001:DB8:1:10::10
Prefix: 64
Gateway: (R1 IPv6 address)
!!!Test
http://2001:DB8:1:30::30 (Should Work)
ping 2001:DB8:1:30::30 (Should Fail after Part 2)

!!! PC2 > Desktop > IP Config

IPv6 Address: 2001:DB8:1:11::20
Prefix: 64
Gateway: (R1 IPv6 address)
!!!Test
http://2001:DB8:1:30::30 (Blocked)
ping 2001:DB8:1:30::30 (Fail after Part 2)

Server3 → Desktop → IP Configuration
IPv6 Address: 2001:DB8:1:30::30
Prefix: 64
Gateway: (R3 IPv6 address)

Desktop → Services → HTTP
Turn ON: HTTP & HTTPS

PRACTICAL 6 - Zone Based Policy Firewall (ZPF)

!!! Script for R3!!!

```
enable
conf t
license boot module c1900 technology-package securityk9
end
write memory
reload

enable
conf t

zone security IN-ZONE
exit
zone security OUT-ZONE
exit

access-list 101 permit ip 192.168.3.0 0.0.0.255 any

class-map type inspect match-all IN-NET-CLASS-MAP
 match access-group 101
exit

policy-map type inspect IN-2-OUT-PMAP
 class type inspect IN-NET-CLASS-MAP
  inspect
 exit
exit

zone-pair security IN-2-OUT-ZPAIR source IN-ZONE destination OUT-ZONE
 service-policy type inspect IN-2-OUT-PMAP
exit

interface g0/1
 zone-member security IN-ZONE
exit

interface s0/0/1
 zone-member security OUT-ZONE
exit>end
```

PC-C (Internal Host – 192.168.3.3) Desktop → Command Prompt

```
ping 192.168.1.3
ssh -l Admin 10.2.2.2
Adminpa55
Exit
```

PC-A (External Server – 192.168.1.3) Desktop → Command Prompt & R2 CLI

```
ping 192.168.3.3
```

Verification on R3 (While SSH/HTTP Active)

```
show policy-map type inspect zone-pair sessions
```

PRACTICAL 10 - IOS INTRUSION DETECTION/ PREVENTION SYSTEM

!!!Script for R1!!!

```
enable
conf t
license boot module c1900 technology-package securityk9
end
write memory
reload

enable
clock set 10:20:00 10 january 2014
mkdir ipsdir
conf t

ip ips config location flash:ipsdir
ip ips name iosips
ip ips notify log
service timestamps log datetime msec
logging host 192.168.1.50

ip ips signature-category
 category all
  retired true
 exit
 category ios_ips basic
  retired false
 exit
exit

interface g0/1
 ip ips iosips out
exit

ip ips signature-definition
 signature 2004 0
  status
   retired false
   enabled true
  exit
  engine
   event-action produce-alert
   event-action deny-packet-inline
  exit
 exit
exit>end
```

From PC-C → Ping PC-A: ping 192.168.1.x
From PC-A → Ping PC-C: ping 192.168.3.x
On R1 : show ip ips all & show ip ips statistics

PRACTICAL 7 - LAYER 2 SECURITY

!!! CENTRAL – Root Bridge
```
enable
conf t
spanning-tree vlan 1 root primary
end
```
!!! SW-1 – Secondary Root + Root Guard
```
enable
conf t
spanning-tree vlan 1 root secondary
interface range f0/23 - 24
spanning-tree guard root
end
```
!!! SW-2 – Root Guard
```
enable
conf t
interface range f0/23 - 24
spanning-tree guard root > end
```
!!! SW-A – PortFast + BPDU Guard + Port Security + Shutdown Unused
```
enable
conf t
interface range f0/1 - 4
spanning-tree portfast
spanning-tree bpduguard enable
exit
interface range f0/1 - 22
switchport mode access
switchport port-security
switchport port-security maximum 2
switchport port-security violation shutdown
switchport port-security mac-address sticky
exit
interface range f0/5 - 22
shutdown
end
```
!!! SW-B – PortFast + BPDU Guard + Port Security + Shutdown Unused
```
enable
conf t
interface range f0/1 - 4
spanning-tree portfast
spanning-tree bpduguard enable
exit
interface range f0/1 - 22
switchport mode access
switchport port-security
switchport port-security maximum 2
switchport port-security violation shutdown
switchport port-security mac-address sticky
exit
interface range f0/5 - 22
Shutdown > end
```

PRACTICAL 9 -

!!! SW-1 – Trunk + VLAN 20 + SVI

```
enable
conf t
interface f0/23
switchport mode trunk
switchport trunk native vlan 15
switchport nonegotiate
no shutdown
exit
vlan 20
exit
interface vlan 20
ip address 192.168.20.3 255.255.255.0
no shutdown > end
```

!!! SW-2 – Trunk + VLAN 20 + SVI

```
enable
conf t
interface f0/23
switchport mode trunk
switchport trunk native vlan 15
switchport nonegotiate
no shutdown
exit
vlan 20
exit
interface vlan 20
ip address 192.168.20.4 255.255.255.0
no shutdown > end
```

!!! SW-A – Create VLAN 20 + Assign PC Port

```
enable
conf t
vlan 20
exit
interface vlan 20
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
interface f0/1
switchport mode access
switchport access vlan 20
no shutdown > end
```

!!! SW-B – VLAN 20 + SVI

```
enable
conf t
vlan 20
exit
```

```
interface vlan 20
ip address 192.168.20.2 255.255.255.0
no shutdown > end
```

!!! CENTRAL – VLAN 20 + SVI

```
enable
conf t
vlan 20
exit
interface vlan 20
ip address 192.168.20.5 255.255.255.0
no shutdown
end
```

!!! R1 – Subinterface + ACL Security

```
enable
conf t

interface g0/0.3
encapsulation dot1q 20
ip address 192.168.20.100 255.255.255.0
exit
access-list 101 deny ip any 192.168.20.0 0.0.0.255
access-list 101 permit ip any any
access-list 102 permit ip host 192.168.20.50 any
interface g0/0.1
ip access-group 101 in
exit
interface g0/0.2
ip access-group 101 in
exit
line vty 0 4
access-class 102 in
end
```

MANAGEMENT PC SETTINGS

Desktop → IP Configuration:

IP Address: 192.168.20.50

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.20.100

PRACTICAL 8 - Conf & VERIFY SITE-To-SITE IPsec VPN using CLI

!!! R1 – COMPLETE IPsec VPN CONFIG

```
enable
conf t
license boot module c1900 technology-package securityk9
end
write memory
reload

enable
conf t

access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255

crypto isakmp policy 10
encryption aes 256
authentication pre-share
group 5
exit

crypto isakmp key vpnpa55 address 10.2.2.2

crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac

crypto map VPN-MAP 10 ipsec-isakmp
description VPN connection to R3
set peer 10.2.2.2
set transform-set VPN-SET
match address 110
exit

interface s0/0/0
crypto map VPN-MAP
end
write memory
```

!!! R3 – COMPLETE IPsec VPN CONFIG

```
enable
conf t
license boot module c1900 technology-package securityk9
end
write memory
reload

enable
conf t
```

```
access-list 110 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255

crypto isakmp policy 10
encryption aes 256
authentication pre-share
group 5
exit

crypto isakmp key vpnpa55 address 10.1.1.2

crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac

crypto map VPN-MAP 10 ipsec-isakmp
description VPN connection to R1
set peer 10.1.1.2
set transform-set VPN-SET
match address 110
exit

interface s0/0/1
crypto map VPN-MAP
end
```

VERIFICATION COMMANDS (Perform on R1)

Before traffic: show crypto ipsec sa

Packets should be **0**.

Generate interesting traffic: From PC-A ping PC-C > ping 192.168.3.x

---

Check again on R1:

show crypto ipsec sa

✔ Encapsulated
✔ Encrypted
✔ Decapsulated
✔ Decrypted

Values should now be **greater than 0**

PRACTICAL 9 - CONF ASA BASIC SETTINGS & FIREWALL USING CLI

```
enable
<Press Enter>
conf t
hostname CCNAS-ASA
domain-name ccnasecurity.com
enable password ciscoenpa55
clock set 13:52:51 June 10 2015
! -------------------------
! INSIDE INTERFACE
! -------------------------
interface vlan 1
nameif inside
ip address 192.168.1.1 255.255.255.0
security-level 100
no shutdown
! -------------------------
! OUTSIDE INTERFACE
! -------------------------
interface vlan 2
nameif outside
ip address 209.165.200.226 255.255.255.248
security-level 0
no shutdown
! -------------------------
! DEFAULT ROUTE
! -------------------------
route outside 0.0.0.0 0.0.0.0 209.165.200.225
! -------------------------
! PAT (INSIDE TO OUTSIDE)
! -------------------------
object network inside-net
subnet 192.168.1.0 255.255.255.0
nat (inside,outside) dynamic interface
exit
! -------------------------
! ICMP INSPECTION (MPF)
! -------------------------
class-map inspection_default
match default-inspection-traffic
exit
policy-map global_policy
class inspection_default
inspect icmp
exit
service-policy global_policy global
! -------------------------
! DHCP CONFIG
! -------------------------
dhcpd address 192.168.1.5-192.168.1.36 inside
dhcpd dns 209.165.201.2 interface inside
```

```
dhcpd enable inside
! -------------------------
! AAA LOCAL AUTH
! -------------------------
username admin password adminpa55
aaa authentication ssh console LOCAL
! -------------------------
! SSH CONFIG
! -------------------------
crypto key generate rsa modulus 1024
<When asked to replace keys type NO>
ssh 192.168.1.0 255.255.255.0 inside
ssh 172.16.3.3 255.255.255.255 outside
ssh timeout 10
! -------------------------
! DMZ CONFIG
! -------------------------
interface vlan 3
nameif dmz
ip address 192.168.2.1 255.255.255.0
security-level 70
no forward interface vlan 1
no shutdown
interface ethernet0/2
switchport access vlan 3
no shutdown
! -------------------------
! STATIC NAT FOR DMZ SERVER
! -------------------------
object network dmz-server
host 192.168.2.3
nat (dmz,outside) static 209.165.200.227
exit
! -------------------------
! OUTSIDE ACL FOR DMZ
! -------------------------
access-list OUTSIDE-DMZ permit icmp any host 192.168.2.3
access-list OUTSIDE-DMZ permit tcp any host 192.168.2.3 eq 80
access-group OUTSIDE-DMZ in interface outside
end
write memory
```